P.9 Combinations of Functions: Composite Functions

Name _____ Period ____ Date _____

Pre-Calculus P.9 (30-minute plan) · Student sheet

I can…

- add, subtract, multiply, and divide functions.
- find and evaluate compositions of functions.
- find the domain of a composition and decompose a function.

Vocabulary · fill in the missing word

Composition of functions — $(f \circ g)(x) =$ _____

Arithmetic combinations — _____

Part 1 · Notes and examples (fill in each note, then work the example)

Arithmetic combinations: $(f + g)(x) = f(x) + g(x)$, and so on; for f/g, also need $g(x)$ _____.

Example 1

$f(x) = 2x + 1$ and $g(x) = x^2 + 2x - 1$. Find $(f + g)(x)$, $(f + g)(3)$, and $(f - g)(2)$.

Composition: $(f \circ g)(x) =$ _____: work from the inside out.

Example 2

$f(x) = x + 2$ and $g(x) = 4 - x^2$. Find $(f \circ g)(x)$, $(g \circ f)(x)$, and $(f \circ g)(-2)$.

Domain of $f \circ g$: x must be in the domain of g, and $g(x)$ must be in the domain of _____.

Example 3

$f(x) = x^2 - 9$ and $g(x) = \sqrt{9 - x^2}$. Find $(f \circ g)(x)$ and its domain.

Part 2 · Partner practice

Partner practice

$f(x) = \sqrt{x}$ and $g(x) = x + 7$. Find $(f \circ g)(2)$.